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WEEK-7 C PROGRAMS

PROGRAM-1 FOR CONDITION 1

CODE

```c
#include <stdio.h>

int main()
{
    int n, find, pos = -1;

    printf("Enter number of elements:\n");
    scanf("%d", &n);

    int arr[n];

    printf("Enter elements:\n");
    for(int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("Enter element to search:\n");
    scanf("%d", &find);

    for(int i = 0; i < n; i++)
    {
        if(arr[i] == find)
        {
            pos = i + 1;
            break;
        }
    }

    if(pos != -1)
        printf("Element found at position %d\n", pos);
    else
        printf("Element not found\n");

    return 0;
}
```

OUTPUT

```
Enter number of elements:
5
Enter elements:
10 20 30 40 50
Enter element to search:
40
Element found at position 4
PS C:\Users\Chethan S\Documents\WEEK-7>
```

PROGRAM-1 FOR CONDITION 2

CODE

```c
#include <stdio.h>

void linear_search(int arr[], int n, int find)
{
    int count = 0;

    for(int i = 0; i < n; i++)
    {
        if(arr[i] == find)
        {
            printf("Element found at position %d\n", i + 1);
            count++;
        }
    }

    if(count == 0)
        printf("Element not found\n");
    else if(count > 1)
        printf("Element occurs %d times\n", count);
}

int main()
{
    int n, find;

    printf("Enter number of elements:\n");
    scanf("%d", &n);

    int arr[n];

    printf("Enter elements:\n");
    for(int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("Enter element to search:\n");
    scanf("%d", &find);

    linear_search(arr, n, find);

    return 0;
}
```

OUTPUT

```
Enter number of elements:
10
Enter elements:
10 30 10 40 50 50 50 30 50 30
Enter element to search:
50
Element found at position 5
Element found at position 6
Element found at position 7
Element found at position 9
Element occurs 4 times
PS C:\Users\Chethan S\Documents\WEEK-7>
```

PROGRAM-1 FOR CONDITION 3

CODE

```c
#include <stdio.h>

void linear_search(int *ptr, int n, int find)
{
    int count = 0;

    for(int i = 0; i < n; i++)
    {
        if(*(ptr + i) == find)
        {
            printf("Element found at position %d\n", i + 1);
            count++;
        }
    }

    if(count == 0)
        printf("Element not found\n");
    else if(count > 1)
        printf("Element occurs %d times\n", count);
}

int main()
{
    int n, find;

    printf("Enter number of elements:\n");
    scanf("%d", &n);

    int arr[n];
    int *ptr = arr;

    printf("Enter elements:\n");
    for(int i = 0; i < n; i++)
        scanf("%d", ptr + i);

    printf("Enter element to search:\n");
    scanf("%d", &find);

    linear_search(ptr, n, find);

    return 0;
}
```

OUTPUT

```
Enter number of elements:
10
Enter elements:
10 20 30 10 10 50 10 50 10 30
Enter element to search:
10
Element found at position 1
Element found at position 4
Element found at position 5
Element found at position 7
Element found at position 9
Element occurs 5 times
PS C:\Users\Chethan S\Documents\WEEK-7>
```

PROGRAM-2

CODE

```c
#include<stdio.h>

int binary_search(int a[],int low,int high,int find)
{
    int mid;

    if(low>high)
        return -1;
    mid=(low+high)/2;

    if(a[mid]==find)
        return mid;

    else if(find<a[mid])
        return binary_search(a,low,mid-1,find);
    else
        return binary_search(a,mid+1,high,find);
}

int main()
{
    int n;
    printf("Enter the number of elements:\n");
    scanf("%d",&n);

    int numbers[n],find,pos;

    printf("Enter the array elements in ascending order:\n");

    for(int i=0;i<n;i++)
        scanf("%d",&numbers[i]);

    printf("\nEnter the element to search:\n");
    scanf("%d",&find);

    pos=binary_search(numbers,0,n-1,find);

    if(pos==-1)
        printf("Element not found\n");

    else
        printf("The element %d is found at position %d\n",find,pos+1);

    return 0;
}
```

OUTPUT

```
{ .\reccursion_binary_search }
Enter the number of elements:
6
Enter the array elements in ascending order:
10 20 30 40 50 60

Enter the element to search:
50
The element 50 is found at position 5
```

PROGRAM-3

CODE

```c
#include<stdio.h>

int main()
{
    int n;
    printf("Enter the number of elements:\n");
    scanf("%d",&n);

    int array[n],min=0,temp;

    printf("Enter the array elements:\n");
    for(int i=0;i<n;i++)
        scanf("%d",&array[i]);

    printf("\nArray before sorting:\n");
    for(int i=0;i<n;i++)
        printf("%d ",array[i]);

    for(int i=0;i<n-1;i++)
    {
        for(int j=i+1;j<n;j++)
        {
            if(array[j]<array[min])
                min=j;
        }
        temp=array[i];
        array[i]=array[min];
        array[min]=temp;
    }

    printf("\n\nArray after sorting in ascending order:\n");

    for(int i=0;i<n;i++)
        printf("%d ",array[i]);

    return 0;
}
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\WEEK-7> cd "c:
Enter the number of elements:
10
Enter the array elements:
2 3 1 4 5 3 8 9 10 1

Array before sorting:
2 3 1 4 5 3 8 9 10 1

Array after sorting in ascending order:
1 1 3 2 3 4 5 8 9 10
```

PROGRAM-4

CODE

```c
#include<stdio.h>

int total_recharge(int amount,int months)
{
    if(months==0)
        return 0;
    return amount+total_recharge(amount,months-1);
}

int main()
{
    int amt,months,total;

    printf("Enter the monthly recharge amount:\n");
    scanf("%d",&amt);

    printf("Enter the number of months:\n");
    scanf("%d",&months);

    total = total_recharge(amt,months);
    printf("Total recharge amount for %d months = %d\n", months,total);
    return 0;
}
```

OUTPUT

```
Enter the monthly recharge amount:
349
Enter the number of months:
6
Total recharge amount for 6 months = 2094
```

PROGRAM-5

CODE

```c
#include<stdio.h>

int total_items(int *ptr,int n)
{
    int sum=0;
    for(int i=0;i<n;i++)
        sum = sum + *(ptr+i);
    return sum;
}

int highest_items(int *ptr,int n)
{
    int max = *ptr;
    for(int i=1;i<n;i++)
    {
        if(*(ptr+i) > max)
            max = *(ptr+i);
    }
    return max;
}

int lowest_items(int *ptr,int n)
{
    int min= *ptr;
    for(int i=1;i<n;i++)
    {
        if(*(ptr+i) < min)
            min = *(ptr+i);
    }
    return min;
}

int main()
{
    int n;
    printf("Enter the number of billing counters:\n");
    scanf("%d",&n);
    int billing_counters[n],total,high,low;

    printf("Enter the number of items sold at each counter:\n");
    for(int i=0;i<n;i++)
        scanf("%d",&billing_counters[i]);

    total = total_items(billing_counters,n);
    high = highest_items(billing_counters,n);
    low = lowest_items(billing_counters,n);

    printf("\nSales Analysis Report\n");
    printf("Total items sold: %d\n",total);
    printf("Highest items sold: %d\n",high);
```

OUTPUT

```
PS C:\Users\Chethan S\Documents\WEEK-7> cd "c:\Users\
; if ($?) { gcc supermarket.c -o supermarket } ; if (
Enter the number of billing counters:
5
Enter the number of items sold at each counter:
120 95 150 80 110

Sales Analysis Report
Total items sold: 555
Highest items sold: 150
Lowest items sold: 80
```